

Question 1 (15 Marks)**Marks**

- a) Find the modulus and argument of the complex numbers w and z **4**

Where $z = \frac{1+i}{1-i}$ and $w = \frac{\sqrt{2}}{1-i}$

- b) Plot the points z , w , $z + w$ from part a) on an accurate **3**

Argand diagram and hence find the exact value of $\tan(\frac{3\pi}{8})$.

- c) The vertices of a square taken anticlockwise are P, Q, R and S. **2**

If the points P and Q are represented by the complex numbers

$z_P = -1 + 4i$ and $z_Q = -3$

Find the other corners of the square R and S and its centre in the form $a+ib$.

- d) Determine the greatest and least values of $\arg z$, **2**

when $|z - 8i - 5| = 6$, answer to the nearest minute.

- e) In the Argand plane **4**

(i) shade: $|z + 3| + |z - 3| \leq 10$ and $3 \leq |z - 3 + 2i| \leq 4$

(ii) sketch: $\arg(z - 5) - \arg(z + 3) = \frac{\pi}{4}$

Question 2 (15 Marks)**Marks**

a) Show $\int_1^2 \frac{1}{x^2} \ln(x+1) dx = \frac{1}{2} \ln \frac{4}{3} + \int_1^2 \frac{1}{x(x+1)} dx$

4

and hence evaluate

$\int_1^2 \frac{1}{x^2} \ln(x+1) dx$ leaving answer in simplest exact form.

b) Simplify $\frac{1}{1-\sin x} - \frac{1}{1+\sin x}$ and

3

hence find

$$\int \frac{1}{1-\sin x} - \frac{1}{1+\sin x} dx$$

c) Find $\int \frac{1}{\sqrt{16-25x^2}} dx$

2

d) Evaluate $\int_0^{\frac{\pi}{4}} \sqrt{1+\cos 4x} dx$ leave answer in exact form.

2

e) Find $\int \frac{-2x+4}{(x^2+1)(x-1)^2} dx$

4

Question 3 (15 Marks)**Marks**

a) Let $t = \tan \frac{\theta}{2}$

(i) Find expressions for $\sin \frac{\theta}{2}$ and $\cos \frac{\theta}{2}$ in terms of t

2

(ii) Hence show $\sin \theta = \frac{2t}{1+t^2}$ and $\cos \theta = \frac{1-t^2}{1+t^2}$

2

(iii) Show that $\frac{dt}{d\theta} = \frac{1}{2}(1+t^2)$

1

(iv) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{5+3\sin \theta + 4\cos \theta}$

4

b) If $I_n = \int x^n (2x+c)^{-\frac{1}{2}} dx$, show that

(i) $I_n = \frac{x^n (2x+c)^{\frac{1}{2}}}{2n+1} - \frac{ncI_{n-1}}{2n+1}$

3

(ii) Hence evaluate $\int_0^1 x^3 (2x+1)^{-\frac{1}{2}} dx$

3

Question 4 (15 Marks)**Marks**

a) The point $P(a\cos\theta, b\sin\theta)$ lies on the ellipse with equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{with } a > b \text{ and eccentricity } e.$$

The foci of the ellipse are S and S' and M , M' are the feet of the perpendiculars from P onto the directrices corresponding to S and S' .

The Normal to the ellipse at P meets the major axis of the ellipse At H .

(i) Draw a sketch to illustrate the above information. 2

(ii) Prove $SP + S'P = 2a$. 1

(iii) Show that the coordinates of H are 3
$$\left(\frac{(a^2 - b^2)\cos\theta}{a}, 0 \right).$$

(iv) Show that $\frac{HS}{HS'} = \frac{1 - e\cos\theta}{1 + e\cos\theta} = \frac{PS}{PS'}$ 2

b) Show that the locus of the point $Q\left\{\frac{a}{2}\left(t + \frac{1}{t}\right), \frac{b}{2}\left(t - \frac{1}{t}\right)\right\}$ for varying 2

values of t is the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

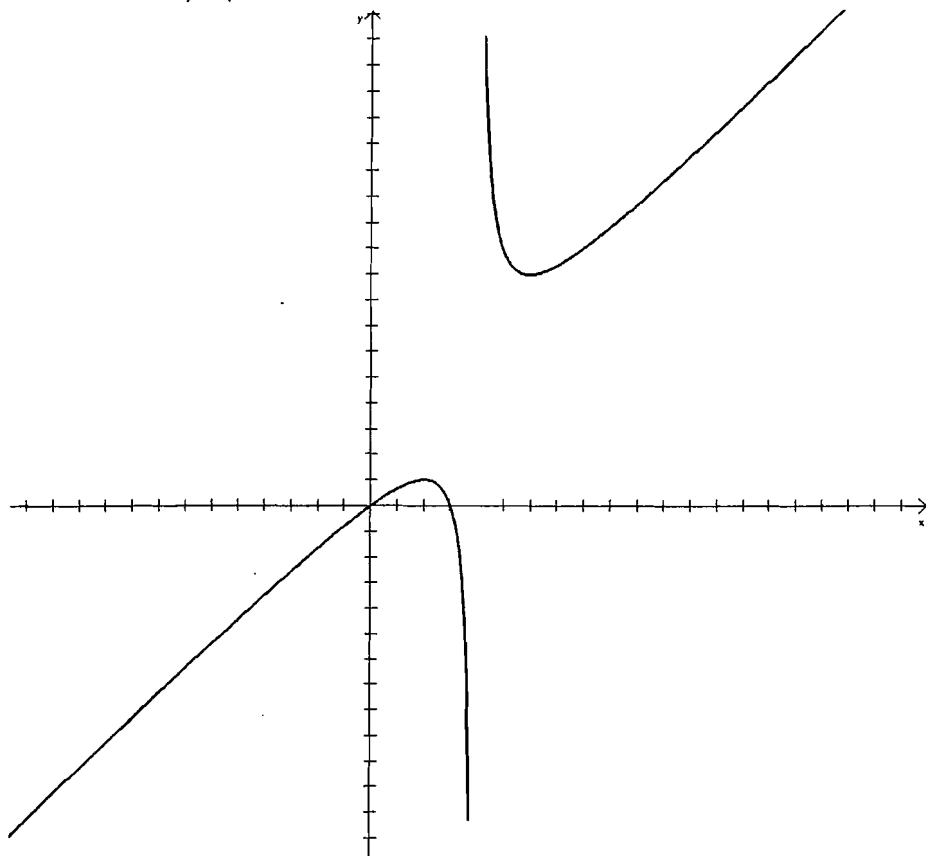
(I) Show the gradient of the tangent at Q is $\frac{b}{a}\left(\frac{t^2 + 1}{t^2 - 1}\right)$ 2

(II) Derive the equation of the tangent at Q 3

Question 5 (15 Marks)

Marks

- a) The diagram shows the graph of $y = f(x)$. The graph has
A vertical asymptote at $x = 4$.



Draw separate one third page sketches of the graphs of the following

(i) $y = \sqrt{f(x)}$ **2**

(ii) $y = \frac{1}{f(x)}$ **2**

(iii) $y^2 = f(x)$ **2**

(iv) $y = \cos(f(x))$ **2**

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Question 5 continued

Marks

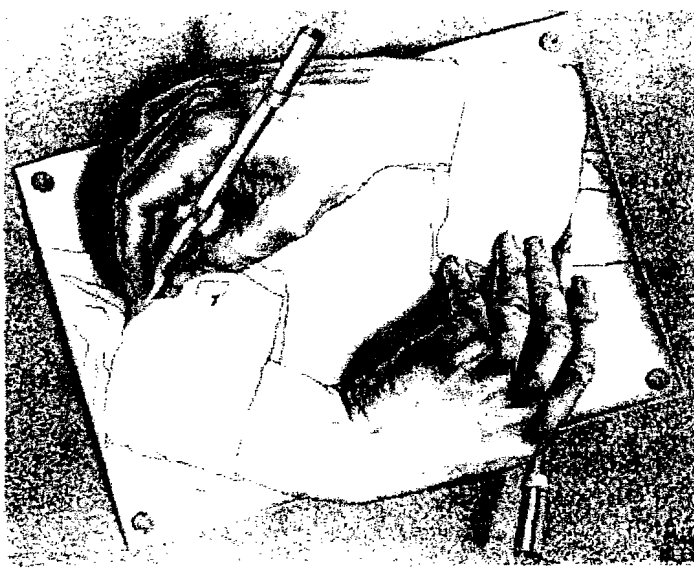
- b) Sketch the graph of $y = x + \frac{x}{x^2 - 25}$
clearly indicating any asymptotes and any points
where the graph meets the axes

4

- c) Find the equation of the normal to the curve
 $x^3y - 3xy^2 + 2y^3 = 6$ at $(1,2)$

3

End of Question 5



Question 6 (15 Marks)**Marks**

- a) If $\frac{p}{q}$ is a zero of the polynomial (p and q are relatively prime)

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0, a_n \neq 0$$

and $a_0, a_1, a_2, a_3, \dots, a_n$ are integers,

- (i) Show q/a_n (q divides a_n) and p/a_0 (p divides a_0) 2

- (ii) Given $P(x) = x^3 - 4x^2 - 3x - 10$ has a rational root, 3
factor $P(x)$ over the complex field.

- b) Show that if the polynomial $P(x)$ has a root of α multiplicity m , 1
then $P'(x)$ has a root of multiplicity $m - 1$.

Given that $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$ has a three fold root, 2
Find all the roots of $P(x)$.

- c) If α, β, δ are the roots of $p(x) = 2x^3 - 4x^2 - 3x - 1$ 2
Find the values of $\alpha^3 + \beta^3 + \delta^3$.

- d) Let $f(t) = t^3 + ct + d$ where c and d are constants
Suppose that the equation $f(t) = 0$ has three distinct real roots
 t_1, t_2 , and t_3 .

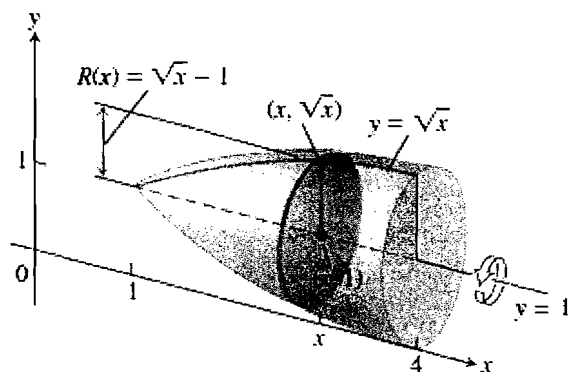
- (i) Show that $t_1^2 + t_2^2 + t_3^2 = -2c$ 2

- (ii) If the function $y = f(t)$ has two turning points at 3
 $t = u$ and $t = v$ and $f(u) \times f(v) < 0$
Show that $27d^2 + 4c^3 < 0$.

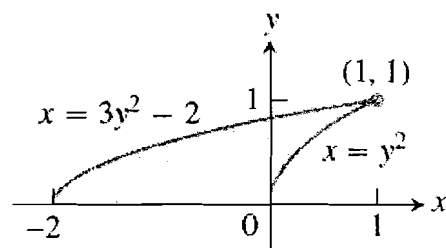
Question 7 (15 Marks)

Marks

- a) Find the volume of the solid generated by revolving the region bounded
By $y = \sqrt{x}$ and the lines $x = 1$, $x = 4$ about the line $y = 1$. **2**
Use the slicing method.



- b) The region shown here is revolved about the x-axis to generate a solid. **3**



Use the method of cylindrical shells to find the volume

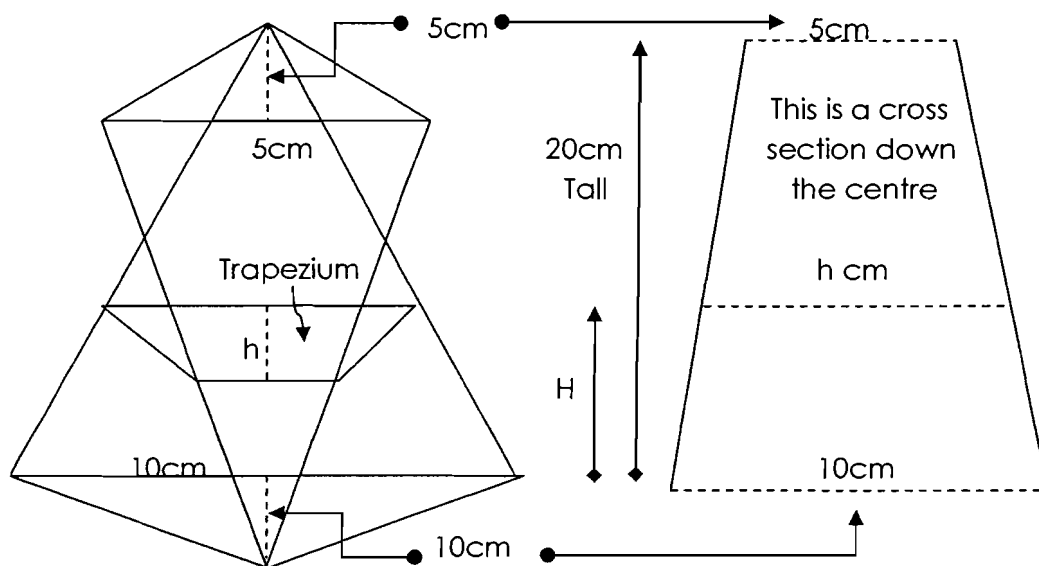
- c) The circle $(x - 6)^2 + (y - 4)^2 = 4$ is rotated around the line $x = 2$. **5**
Calculate the exact volume generated.

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Question 7 continued

Marks

- d) A Saltshaker 20 cm tall is made with isosceles triangular ends and a cross section which is an isosceles trapezium. **5**
 Note top and bottom triangles have bases and perpendicular heights equal.



The Trapezium is located H cm above the base, show using similarity

that the trapezium has an area of $A = 50 - \frac{10H}{4} + \frac{H^2}{32} \text{ cm}^2$

Hence find the volume of the saltshaker to the nearest millilitre.

Question 8 (15 Marks)**Marks**

- a) Use De Moivre's theorem to express $\cos 5\theta$, $\sin 5\theta$
in terms of $\sin \theta$ and $\cos \theta$.

5

Hence express $\tan 5\theta$ as a rational function of t where $t = \tan \theta$.

Deduce that $\tan \frac{\pi}{5} \tan \frac{2\pi}{5} \tan \frac{3\pi}{5} \tan \frac{4\pi}{5} = 5$.

- b) Find $\int \ln(\sqrt{x} + \sqrt{1+x}) dx$

3

- c) If $y = \frac{1}{2}(e^{ax} - e^{-ax})$

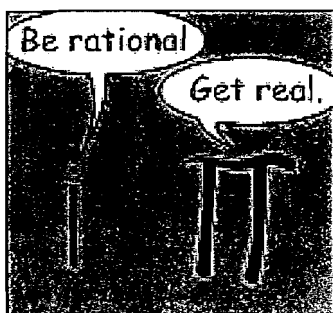
(i) Show that $x = \frac{1}{a} \ln(y + \sqrt{1+y^2})$

2

(ii) Show that $\left(\frac{dy}{dx}\right)^2 - a^2 y^2 = a^2$

2

(iii) Hence deduce that $\int \frac{dy}{\sqrt{1+y^2}} = \log_e(y + \sqrt{1+y^2}) + c$

3**End of Examination**

$$\begin{aligned}
 du &= \frac{1}{1+u} du & V &= \int x^{-2} dx = -\frac{1}{x} \\
 &= \frac{1}{x} \ln(x+1) \Big|_1^2 + \int_1^2 \frac{dx}{x(x+1)} \\
 &= -\frac{1}{2} \ln 3 + \frac{1}{2} \ln 2 + \int_1^2 \frac{dx}{x(x+1)} \quad \text{Now} \\
 \checkmark &= \frac{1}{2} \ln 3 + \frac{1}{2} \ln 4 + \int_1^2 \frac{1}{x} - \frac{1}{x+1} dx & \frac{1}{x(x+1)} &= \frac{A}{x} + \frac{B}{x+1} \\
 &= \frac{1}{2} \ln \frac{4}{3} + \ln \left(\frac{x}{x+1} \right) \Big|_1^2 & 1 &= A(x+1) + B(x) \\
 &= \frac{1}{2} \ln \frac{4}{3} + \ln \frac{2}{3} - \ln \frac{1}{2} & \text{let } x=0 & \therefore A=1 \\
 &= \frac{1}{2} \ln \frac{4}{3} + \ln \frac{4}{3} = \frac{3}{2} \ln \frac{4}{3} & x=-1 & \therefore B=-1
 \end{aligned}$$

$\boxed{\frac{1}{4}}$

$$\begin{aligned}
 b) \quad \frac{1}{1+\sin x} - \frac{1}{1+\sin x} &= \frac{1+\sin x - (1-\sin x)}{1-\sin^2 x} \\
 \checkmark &= \frac{2\sin x}{\cos^2 x} = 2 \cdot (\cos x)^{-1} \frac{\sin x}{\cos x} \\
 &= 2 \sec x \tan x \\
 \therefore \int 2 \sec x \tan x dx &= 2 \sec x + C
 \end{aligned}$$

$\boxed{\frac{1}{3}}$

$$\begin{aligned}
 c) \quad \int \frac{dx}{\sqrt{16-25x^2}} &= \int \frac{dx}{\sqrt{4^2-(5x)^2}} \quad \text{let } u=5x \\
 &= \frac{1}{5} \int \frac{du}{\sqrt{4^2-u^2}} & du &= 5dx \\
 &= \frac{1}{5} \sin^{-1} \left(\frac{u}{4} \right) + C \\
 &= \frac{1}{5} \sin^{-1} \left(\frac{5x}{4} \right) + C
 \end{aligned}$$

$\boxed{\frac{1}{2}}$

$$\begin{aligned}
 d) \quad \int_0^{\frac{\pi}{4}} \sqrt{1+\cos 4x} dx & \quad \text{Note } 1+\cos 4x = 2\cos^2 2x \\
 &= \int_0^{\frac{\pi}{4}} \sqrt{2\cos^2 2x} dx = \sqrt{2} \int_0^{\frac{\pi}{4}} \cos 2x dx \\
 &= \frac{\sqrt{2}}{2} [\sin 2x]_0^{\frac{\pi}{4}} = \frac{\sqrt{2}}{2} [1-0] \\
 &= \frac{\sqrt{2}}{2}
 \end{aligned}$$

$\boxed{\frac{1}{2}}$

$$\begin{aligned}
 e) \quad \frac{-2x+4}{(x^2+1)(x-1)^2} &= \frac{Ax+B}{x^2+1} + \frac{C}{x-1} + \frac{D}{(x-1)^2} \\
 -2x+4 &= (Ax+B)(x-1)^2 + C(x-1)(x^2+1) + D(x^2+1) \\
 \text{let } x=1 & \therefore 2=2D \therefore \boxed{D=1} \\
 \text{equating coeff } x^3 & \therefore A+C=0 \therefore A=-C \\
 x^2 & \therefore -2A+B-C+D=0 \text{ as } D=1 \\
 & \therefore -2A+B-C=-1 \text{ if } A=-C \therefore 2C+B-C=-1 \\
 x & \therefore -2=A-2B+C & C+B &= -1 \\
 \text{if } A=-C & & \therefore C &= -2 \\
 & \therefore -2B=-2 & \therefore B &= 1
 \end{aligned}$$

$$\begin{aligned}
 \int \frac{-2x+4}{(x^2+1)(x-1)^2} dx &= \int \frac{2x+1}{x^2+1} - \frac{2}{x-1} + \frac{1}{(x-1)^2} dx \\
 &= \int \frac{2x}{x^2+1} + \frac{1}{x^2+1} - \frac{2}{x-1} - \frac{1}{x-1} dx \\
 &= \ln(x^2+1) + \tan^{-1} x - 2\ln(x-1) - \frac{1}{x-1} + C
 \end{aligned}$$

$\boxed{\frac{1}{4}}$

$$\begin{aligned} 11) \sin \theta &= \sin\left(\frac{\theta}{2} + \frac{\theta}{2}\right) = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \\ &= 2 \cdot \frac{t}{\sqrt{1+t^2}} \cdot \frac{1}{\sqrt{1+t^2}} = \frac{2t}{1+t^2} \checkmark \end{aligned}$$

$$\begin{aligned} \cos \theta &= \cos\left(\frac{\theta}{2} + \frac{\theta}{2}\right) = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \\ &= \frac{1}{1+t^2} - \frac{t^2}{1+t^2} = \frac{1-t^2}{1+t^2} \checkmark \end{aligned}$$

$$\boxed{\frac{1}{2}}$$

$$t = \tan \frac{\theta}{2} \quad \frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2} = \frac{1}{2} (1 + \tan^2 \frac{\theta}{2})$$

$$\frac{dt}{d\theta} = \frac{1}{2} (1+t^2) \checkmark$$

$$\boxed{\frac{1}{1}}$$

$$\int_0^{\frac{\pi}{2}} \frac{dx}{5+3\sin x + 4\cos x}$$

$$\text{let } t = \tan \frac{x}{2} \therefore dt = \sec^2 \frac{x}{2} \cdot \frac{1}{2} dx$$

$$\sin x = \frac{2t}{1+t^2}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$\int_0^{\frac{\pi}{2}} \frac{dx}{5+3 \cdot \frac{2t}{1+t^2} + 4 \cdot \frac{(1-t^2)}{1+t^2}} = \int_0^1 \frac{2dt}{5(1+t^2) + 6t + 4(1-t^2)}$$

$$\begin{aligned} \int_0^1 \frac{2dt}{9+6t+t^2} &= \int_0^1 \frac{2dt}{(t+3)^2} = 2 \int_0^1 \frac{1}{(t+3)^2} dt \\ &= 2 \left[-\frac{1}{t+3} \right]_0^1 = -2 \left[\frac{1}{4} - \frac{1}{3} \right] = -2 \cdot \frac{-1}{12} \end{aligned}$$

$$\boxed{\frac{1}{6}}$$

$$\boxed{\frac{1}{4}}$$

$$du = n x^{n-1} dx$$

$$dV = (2x+c)^{\frac{1}{2}} dx$$

$$V = \frac{(2x+c)^{\frac{1}{2}}}{2 \times \frac{1}{2}} = (2x+c)^{\frac{1}{2}}$$

$$\therefore I_n = x^n (2x+c)^{\frac{1}{2}} - \int n x^{n-1} (2x+c)^{\frac{1}{2}} dx$$

$$= x^n (2x+c)^{\frac{1}{2}} - \int n x^{n-1} (2x+c)^{\frac{1}{2}} dx$$

$$I_n = x^n (2x+c)^{\frac{1}{2}} - 2n \int x^n (2x+c)^{\frac{1}{2}} dx - n c I_{n-1}$$

$$\therefore (2n+1) I_n = x^n (2x+c)^{\frac{1}{2}} - n c I_{n-1}$$

$$\therefore I_n = \frac{x^n (2x+c)^{\frac{1}{2}}}{2n+1} + \frac{n c I_{n-1}}{2n+1}$$

$$\boxed{\frac{1}{3}}$$

$$\therefore I_3 = \frac{x^3 (2x+c)^{\frac{1}{2}}}{7} \Big|_0^1 - \frac{3 I_2}{7}$$

$$= \frac{\sqrt{3}}{7} - \frac{3}{7} I_2$$

$$I_2 = \frac{x^2 (2x+c)^{\frac{1}{2}}}{5} \Big|_0^1 - \frac{2 I_1}{5}$$

$$= \frac{\sqrt{3}}{5} - \frac{2 I_1}{5}$$

$$I_1 = \frac{x (2x+c)^{\frac{1}{2}}}{3} \Big|_0^1 - \frac{I_0}{3} = \frac{\sqrt{3}}{3} - \frac{I_0}{3}$$

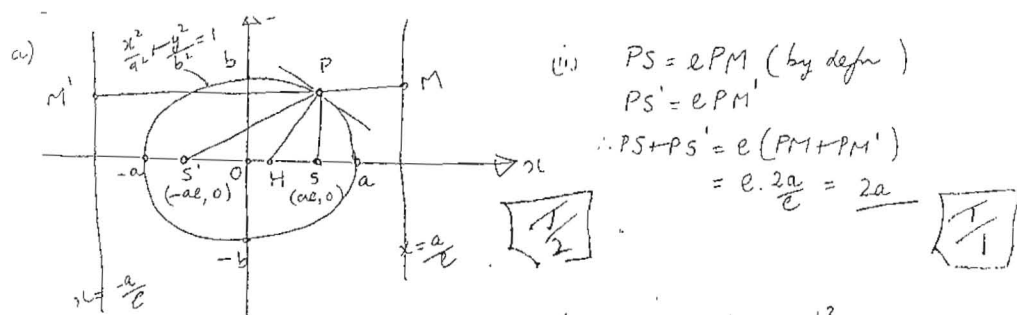
$$I_0 = \int_0^1 (2x+c)^{-\frac{1}{2}} dx = \frac{(2x+c)^{\frac{1}{2}}}{2 \times \frac{1}{2}} \Big|_0^1 = \sqrt{3} - 1$$

$$\therefore I_3 = \frac{\sqrt{3}}{7} - \frac{3}{7} \left[\frac{\sqrt{3}}{5} - \frac{2}{5} \left[\frac{\sqrt{3}}{3} - \frac{1}{3} (\sqrt{3} - 1) \right] \right]$$

$$= \frac{\sqrt{3}}{7} - \frac{3}{7} \left[\frac{\sqrt{3}}{5} - \frac{2}{5} \cdot \frac{1}{3} \right]$$

$$= \frac{\sqrt{3}}{7} - \frac{3}{7} \left[\frac{3\sqrt{3}-2}{5} \right] = \frac{5\sqrt{3}-3\sqrt{3}+2}{35}$$

$$= \frac{2\sqrt{3}+2}{35}$$



(iii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\therefore \frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0 \therefore \frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$

at $P(a \cos \theta, b \sin \theta) \therefore \frac{dy}{dx} = -\frac{b^2 a \cos \theta}{a^2 b \sin \theta} = -\frac{b}{a} \cot \theta$

$\therefore$ Gradient of normal $m = \frac{a}{b} \tan \theta$ using $m_1 m_2 = -1$

Eqⁿ of normal $\frac{y - b \sin \theta}{x - a \cos \theta} = \frac{a}{b} \tan \theta$
 $\therefore y = \frac{a}{b} \tan \theta (x - a \cos \theta) + b \sin \theta$
 $y = \frac{a}{b} x \tan \theta - \frac{a^2}{b} \sin \theta + b \sin \theta$

When $y = 0$ $\sin \theta \left(\frac{a^2 - b^2}{b} \right) = \frac{a^2}{b} \tan \theta$
 $\therefore x = \frac{a^2 - b^2}{a} \cdot \frac{\sin \theta \cos \theta}{\sin \theta} = \frac{a^2 - b^2}{a} \cos \theta$

$\therefore H \left(\frac{a^2 - b^2}{a} \cos \theta, 0 \right)$ I

(iv) $\frac{HS}{HS'} = \frac{OS - OH}{OS' + OH} = \frac{ae - \frac{a^2 - b^2}{a} \cos \theta}{ae + \frac{a^2 - b^2}{a} \cos \theta}$ using $b^2 = a^2(1 - e^2)$

$= \frac{a^2 e - (a^2 - b^2) \cos \theta}{a^2 e + (a^2 - b^2) \cos \theta} = \frac{a^2 e - (a^2 - a^2(1 - e^2)) \cos \theta}{a^2 e + (a^2 - a^2(1 - e^2)) \cos \theta}$

$= \frac{a^2 e - a^2 e^2 \cos \theta}{a^2 e + a^2 e^2 \cos \theta} = \frac{a^2 e (1 - e \cos \theta)}{a^2 e (1 + e \cos \theta)}$

$= \frac{1 - e \cos \theta}{1 + e \cos \theta}$

$\frac{PS}{PS'} = \frac{ePM}{ePM'} = \frac{PM}{PM'} = \frac{\frac{a}{e} - a \cos \theta}{\frac{a}{e} + a \cos \theta} = \frac{a - ae \cos \theta}{a + ae \cos \theta}$

$= \frac{a(1 - e \cos \theta)}{a(1 + e \cos \theta)} = \frac{1 - e \cos \theta}{1 + e \cos \theta} = \frac{HS}{HS'}$ I

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ $x^2 = \frac{a^2}{4} \left(t + \frac{1}{t} \right)^2$ $y^2 = \frac{b^2}{4} \left(t - \frac{1}{t} \right)^2$

LHS $\frac{1}{4} \left(t + \frac{1}{t} \right)^2 - \frac{1}{4} \left(t - \frac{1}{t} \right)^2 = \frac{1}{4} \left(t^2 + 2 + \frac{1}{t^2} \right) - \frac{1}{4} \left(t^2 - 2 + \frac{1}{t^2} \right)$
 $= \frac{1}{4} + \frac{1}{4} = 1 = RHS$ I

Q lies on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \therefore \frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 1$

$\therefore \frac{dy}{dx} = \frac{x b^2}{y a^2} = \frac{\frac{b^2 a}{2} \left(t + \frac{1}{t} \right)}{\frac{a^2 b}{2} \left(t - \frac{1}{t} \right)} = \frac{a b^2 \left(\frac{t^2 + 1}{t} \right)}{a^2 b \left(\frac{t^2 - 1}{t} \right)} = \frac{b}{a} \left(\frac{t^2 + 1}{t^2 - 1} \right)$

(ii) Equation of tangent at Q I

$y - \frac{b}{2} \left(\frac{t^2 - 1}{t} \right) = \frac{b}{a} \left(\frac{t^2 + 1}{t^2 - 1} \right) \left(x - \frac{a}{2} \left(\frac{t^2 + 1}{t} \right) \right)$
 $= \frac{b x}{a} \left(\frac{t^2 + 1}{t^2 - 1} \right) - \frac{b}{2} \left(\frac{(t^2 + 1)^2}{t(t^2 - 1)} \right)$

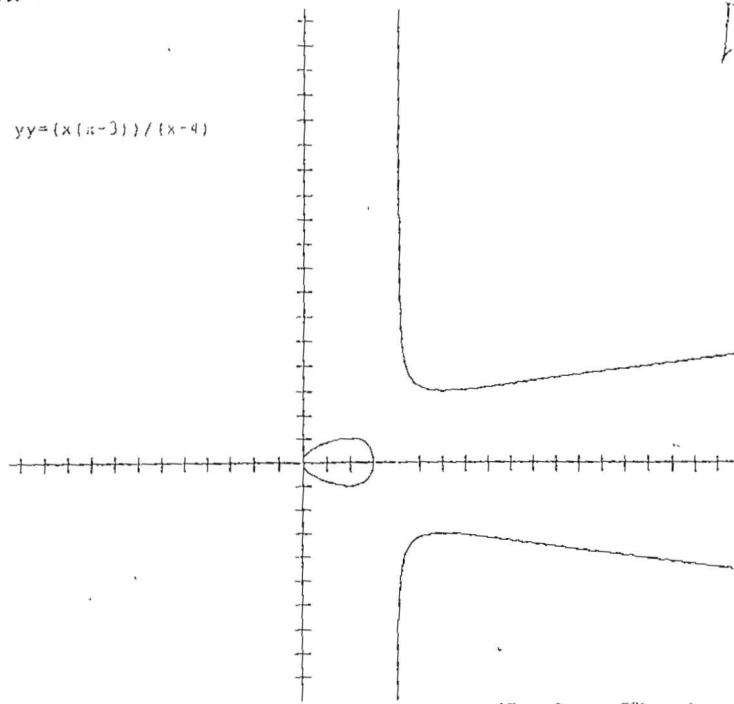
$\therefore y - \frac{b x}{a} \left(\frac{t^2 + 1}{t^2 - 1} \right) = \frac{b}{2} \left[\frac{t^2 - 1}{t} - \frac{(t^2 + 1)^2}{t(t^2 - 1)} \right]$
 $= \frac{b}{2} \left[\frac{(t^2 - 1)^2 - (t^2 + 1)^2}{t(t^2 - 1)} \right]$
 $= \frac{b}{2} \left[\frac{(t^2 - 1 + t^2 + 1)(t^2 - 1 - t^2 - 1)}{t(t^2 - 1)} \right]$

$\therefore y - \frac{b x}{a} \left(\frac{t^2 + 1}{t^2 - 1} \right) = \frac{b}{2} \cdot \frac{2t^2 x - 2}{t(t^2 - 1)} = \frac{-2bt}{t^2 - 1}$

$\therefore \frac{b x}{a} \left(\frac{t^2 + 1}{t^2 - 1} \right) - (t^2 - 1) y - 2bt = 0$ I

5. ii)

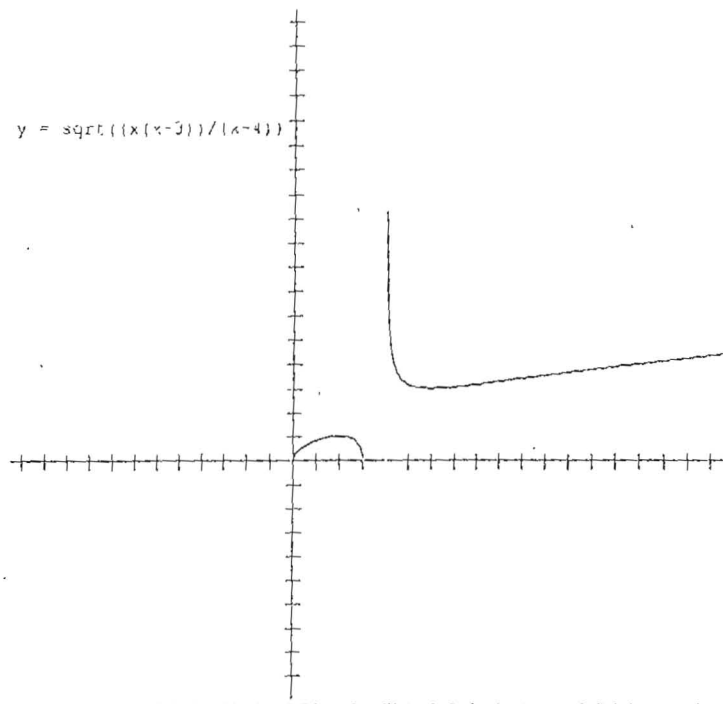
$$y = (x(x-3))/(x-4)$$



2 each

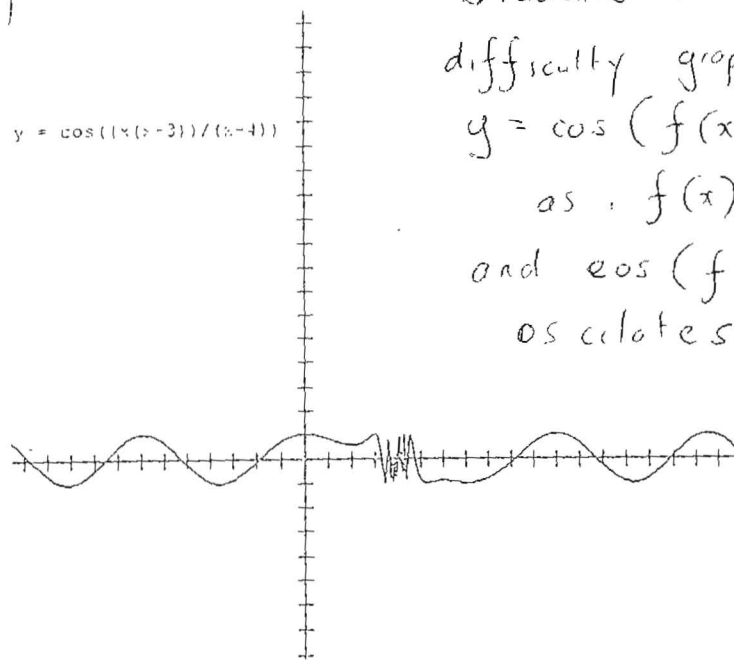
i)

$$y = \sqrt{(x(x-3))/(x-4)}$$



iv)

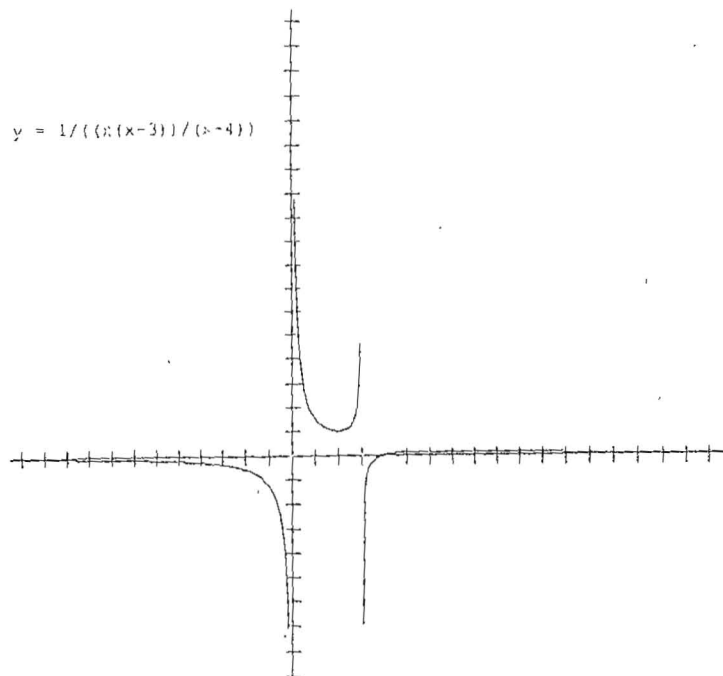
$$y = \cos((x(x-3))/(x-4))$$



Students had
difficulty graphing
 $y = \cos(f(x))$
as $f(x) \rightarrow \infty$
and $\cos(f(x))$
oscillates faster

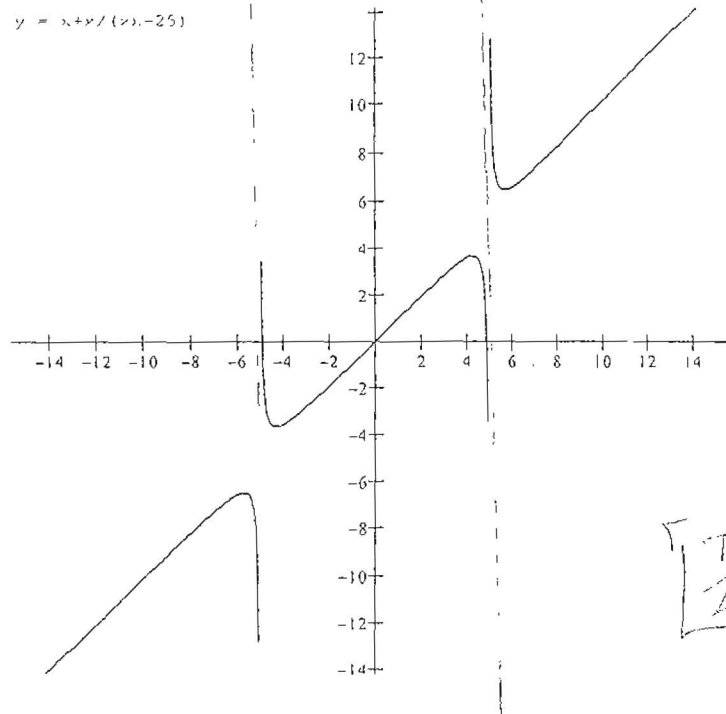
ii)

$$y = 1/((x(x-3))/(x-4))$$



Answer Q5 b)

$$y = x + x/(x^2 - 25)$$



$\frac{T}{4}$

Suggested Marking
asymptotes $y = x$ $x = \pm 5$
analysis $\lim_{x \rightarrow \pm 5^{\pm}}$

Many students had difficulty
determining the precise nature
of the function
for $-5 < x < 5$

Q5 a) $x^3y - 3xy^2 + 2y^3 = 6$

$$x^3 \frac{dy}{dx} + y 3x^2 - 3[x \cdot 2y \frac{dy}{dx} + y^2 \cdot 1] + 2 \cdot 3y^2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} [x^3 - 6xy + 6y^2] + 3x^2y - 3y^3 = 0$$

$$\therefore \frac{dy}{dx} = \frac{3y^2 - 3x^2y}{x^3 - 6xy + 6y^2}$$

at $(1, 2)$

$$\frac{dy}{dx} = \frac{3 \cdot 4 - 3 \cdot 1 \cdot 2}{1 - 6 \cdot 2 + 24}$$

$$\frac{dy}{dx} = \frac{12 - 6}{1 - 12 + 24} = \frac{6}{13}$$

$\therefore$ eqn of normal. $m = -\frac{13}{6}$ * errors in finding gradient of tangent

$$y - 2 = -\frac{13}{6}(x - 1)$$

$$6y - 12 = -13x + 13$$

$$13x + 6y - 25 = 0$$

eqn of normal.

$\frac{T}{3}$

$\frac{CT}{15}$

* sometimes gradient of tangent was used instead of gradient of normal

show 6

(i) $\frac{P}{q}$ is root of $P(x)$

$$a_n \left(\frac{p}{q}\right)^n + a_{n-1} \left(\frac{p}{q}\right)^{n-1} + \dots + a_1 \frac{p}{q} + a_0 = 0 \quad \times q^n$$

$$a_n p^n + a_{n-1} p^{n-1} q + \dots + a_1 p q^{n-1} + a_0 q^n = 0$$

$$-a_n p^n = a_{n-1} p^{n-1} q + \dots + a_1 p q^{n-1} + a_0 q^n$$

$$= q(a_{n-1} p^{n-1} + \dots + a_1 p q^{n-2} + a_0 q^{n-1})$$

Note any exponents with $\frac{p}{q}$ is fractional

Since p, q relatively prime so q and p^n are also relatively prime
 $\therefore q/a_n$ (q divides a_n)

Similarly $-a_0 q^n = a_n p^n + a_{n-1} p^{n-1} q + \dots + a_1 p q^{n-1}$
 $= p(a_n p^{n-1} + a_{n-1} p^{n-2} q + \dots + a_1 q^{n-1})$

Since p, q rel prime p, q^n rel. prime
 p/a_0

(ii) $P(x) = x^5 - 4x^2 - 3x - 10$ monic $a_n = 1$
 $P/-10 \rightarrow \pm 1 \pm 2 \pm 5 \pm 10$
 $P(5) = 0 \quad P(x) = (x-5)(x^2 + x + 2)$
 $= (x-5)(x - \frac{-1+\sqrt{7}i}{2})(x - \frac{-1-\sqrt{7}i}{2})$
 over $\mathbb{C}$

$P(x) = (x-\alpha)^m Q(x)$
 $P'(x) = (x-\alpha)^m Q'(x) + Q(x) \cdot m(x-\alpha)^{m-1}$
 $= (x-\alpha)^{m-1} ((x-\alpha)Q'(x) + mQ(x))$

ie $P'(x)$ has root α of multiplicity $(m-1)$
 Since $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$ 3 fold
 $P'(x) = 4x^3 + 3x^2 - 6x - 5$ 2 fold
 $P''(x) = 12x^2 + 6x - 6$ 1 fold
 $= 6(2x-1)(x+1)$
 $P''(x) = 0$ for $x = \frac{1}{2}, -1$ but $P'(\frac{1}{2}) \neq 0$
 $P'(-1) = 0$
 $\therefore P(x) = (x+1)^3(x-2)$ by inspect
 Hence roots $x = -1, -1, -1, 2$

(c)

$$2\alpha^3 - 4\alpha^2 - 3\alpha - 1 = 0$$

$$2\beta^3 - 4\beta^2 - 3\beta - 1 = 0$$

$$2\delta^3 - 4\delta^2 - 3\delta - 1 = 0$$

$$\therefore 2(\alpha^3 + \beta^3 + \delta^3) = 4(\alpha^2 + \beta^2 + \delta^2) + 3(\alpha + \beta + \delta) + 3$$

$$2(\alpha + \beta + \delta)^2 = 4(\alpha + \beta + \delta)^2 - 2(\alpha\beta + \beta\delta + \delta\alpha) = 2^2 - 2(-\frac{3}{2}) = 7$$

$$\alpha^3 + \beta^3 + \delta^3 = 2 \cdot 7 + \frac{3}{2} \cdot 2 + \frac{3}{2} = 18\frac{1}{2}$$

$$= 14 + 3 + 1\frac{1}{2}$$

$\frac{7}{2}$

d) $t_1^2 + t_2^2 + t_3^2 = (t_1 + t_2 + t_3)^2 - 2(t_1 t_2 + t_2 t_3 + t_1 t_3)$
 but $t_1 t_2 + t_2 t_3 + t_1 t_3 = C \quad \leq \alpha_c = -\frac{b}{a} = 0$

(i) $\therefore t_1^2 + t_2^2 + t_3^2 = 0^2 - 2C = -2C$

$\frac{7}{2}$

(ii) $f(t) = t^3 + ct + d$
 $f(t) = 3t^2 + C = 0 \quad \therefore t = \pm\sqrt{\frac{-C}{3}}$
 let $u = \sqrt{\frac{-C}{3}}, v = -\sqrt{\frac{-C}{3}} \quad f(u)f(v) = (u^3 + cu + d)(v^3 + cv + d)$
 $\therefore f(u)f(v) = (-\frac{C}{3}\sqrt{\frac{-C}{3}} + c\sqrt{\frac{-C}{3}} + d)(\frac{C}{3}\sqrt{\frac{-C}{3}} - c\sqrt{\frac{-C}{3}} + d)$
 simplify first
 $= (\frac{2c}{3}\sqrt{\frac{-C}{3}} + d)(-\frac{2c}{3}\sqrt{\frac{-C}{3}} + d)$ diff of 2 square
 $= d^2 - \frac{4c^2}{9} \cdot \frac{-C}{3} < 0$
 $= d^2 + \frac{4c^3}{27} < 0$
 $\therefore 27d^3 + 4c^3 < 0$

$\frac{7}{3}$

a) $\delta V = \pi r^2 \delta z$ $r(z) = \sqrt{z} - 1$

$$\therefore \delta V = \pi (\sqrt{z} - 1)^2 \delta z \quad \therefore V = \lim_{\delta z \rightarrow 0} \sum_{z=1}^4 \pi (\sqrt{z} - 1)^2 \delta z$$

$$V = \pi \int_1^4 (z - 2\sqrt{z} + 1) dz$$

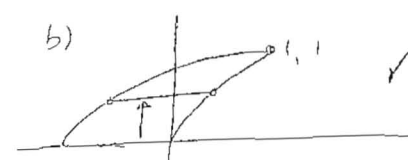
$$= \pi \left[\frac{z^2}{2} - \frac{4}{3} z^{3/2} + z \right]_1^4 = \pi \left[8 - \frac{32}{3} + 4 - \left(\frac{1}{2} - \frac{4}{3} + 1 \right) \right]$$

$$= \pi \left[12 - 10\frac{2}{3} + \left(-\frac{1}{6} \right) \right] = \pi \cdot \frac{1}{6} = \frac{\pi}{6} u^3$$

≈ 5 min

$\frac{1}{2}$

b)



$$\delta V = 2\pi r h \delta y \quad h = y^2 - (3y^2 - 2) = 2 - 2y^2$$

$$\therefore \delta V = 2\pi y (2 - 2y^2) \delta y$$

$$V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^1 2\pi (2y - 2y^3) \delta y$$

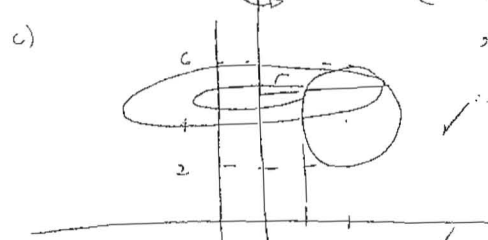
$$= 2\pi \int_0^1 (2y - 2y^3) dy = 4\pi \left[\frac{y^2}{2} - \frac{y^4}{4} \right]_0^1 = 4\pi \left[\frac{1}{2} - \frac{1}{4} \right] = \pi u^3$$

Core needed to be taken in determining volume

< 5 min

$\frac{1}{3}$

c)



$$(x-6)^2 + (y-4)^2 = 4$$

$$x-6 = \pm \sqrt{4 - (y-4)^2}$$

$$x = 6 \pm \sqrt{4 - (y-4)^2}$$

$$R = 6 + \sqrt{4 - (y-4)^2} - 2 = 4 + \sqrt{4 - (y-4)^2}$$

$$r = 4 - \sqrt{4 - (y-4)^2}$$

$$\delta V = \pi (R^2 - r^2) \delta y$$

$$\delta V = \pi (R+r)(R-r) \delta y$$

$$\delta V = \pi (8) (2\sqrt{4 - (y-4)^2}) \delta y$$

$$V = \lim_{\delta y \rightarrow 0} \sum_{y=2}^6 16\pi \sqrt{4 - (y-4)^2} \delta y$$

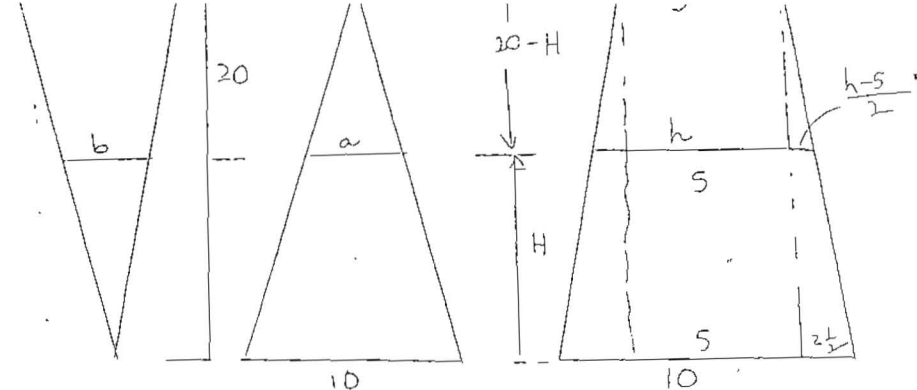
$$V = 16\pi \int_2^6 \sqrt{4 - (y-4)^2} dy$$

$$= 16\pi \cdot 2\pi$$

$$= 32\pi^2 u^3$$

many approaches were taken, after gaps were found in the set-up leading to an incomplete soln

$\frac{1}{5}$



for b $\frac{b}{5} = \frac{H}{20}$

$$b = \frac{H}{4}$$

for a $\frac{a}{10} = \frac{20-H}{20}$

$$20a = 10(20-H)$$

$$a = 10 - \frac{H}{2}$$

for h $\frac{h-5}{5} = \frac{20-H}{20}$

$$h-5 = \frac{20-H}{4}$$

$$h = 10 - \frac{H}{4}$$

$$A = \frac{1}{2} L (a+b)$$

$$A = \frac{1}{2} \left(10 - \frac{H}{4} \right) \left[10 - \frac{H}{2} + \frac{H}{4} \right]$$

$$A = \frac{1}{2} \left(10 - \frac{H}{4} \right) \left(10 - \frac{H}{4} \right)$$

$$= \frac{1}{2} \left(100 - 2 \times 10 \times \frac{H}{4} + \frac{H^2}{16} \right)$$

$$= 50 - \frac{10H}{4} + \frac{H^2}{32}$$

$$V = \int_0^{20} A dH = \int_0^{20} \left(50 - \frac{10H}{4} + \frac{H^2}{32} \right) dH$$

$$= \left[50H - \frac{10H^2}{8} + \frac{H^3}{96} \right]_0^{20}$$

$$= 50 \cdot 20 - \frac{10 \cdot 20^2}{8} + \frac{20^3}{96} - 0$$

$$= 1000 - 500 + 83\frac{1}{3}$$

$$V = 583\frac{1}{3} \text{ cm}^3$$

usually well done if attempted

$\frac{1}{5}$

from 8

$$(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta \quad (\text{by De Moivre's theorem})$$

expand LHS using pascals Δ

$$(\cos \theta)^5 + 5(\cos \theta)^4(i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 + 10 \cos^2 \theta (i \sin \theta)^3 + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5$$

grouping real and imaginary parts

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

$$\tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta} = \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

$$= \cos^5 \theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

$$t = \tan \theta \quad \tan 5\theta = \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4} \quad \text{as a rational function of } t$$

If $\tan 5\theta = 0$ $5\theta = n\pi$ for $n=0, 1, 2, 3, 4$ distinct
 $\therefore \theta = 0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}$

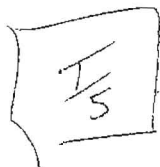
$5t - 10t^3 + t^5 = 0$ need this eqn

$$t(5 - 10t^2 + t^4) = 0$$

$t^4 - 10t^2 + 5 = 0$ has roots $t = \tan \frac{\pi}{5}, \tan \frac{2\pi}{5}, \dots$

The product of roots = $\frac{5}{1} = 5$

$$\tan \frac{\pi}{5} \tan \frac{2\pi}{5} \tan \frac{3\pi}{5} \tan \frac{4\pi}{5} = 5$$



b) $\int \ln(\sqrt{x} + \sqrt{1+x}) dx$ let $u = \ln(\sqrt{x} + \sqrt{1+x})$

$$\frac{du}{dx} = \frac{1}{\sqrt{x} + \sqrt{1+x}} \left(\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{1+x}} \right)$$

$$= \frac{1}{\sqrt{x} + \sqrt{1+x}} \left[\frac{2\sqrt{1+x} + 2\sqrt{x}}{4\sqrt{1+x} \cdot \sqrt{x}} \right]$$

$$= \frac{1}{2\sqrt{x}\sqrt{1+x}}$$

$$dv = dx \quad \therefore v = x$$

By Parts

$$= x \ln(\sqrt{x} + \sqrt{1+x}) - \frac{1}{2} \int \frac{x dx}{\sqrt{x}\sqrt{1+x}}$$

$$= x \ln(\sqrt{x} + \sqrt{1+x}) - \frac{1}{2} \int \frac{x dx}{\sqrt{x^2 + x}}$$

$$x^2 + x = \left(x + \frac{1}{2}\right)^2 - \frac{1}{4}$$

$$= x \ln(\sqrt{x} + \sqrt{1+x}) - \frac{1}{2} \int \frac{x}{\left(x + \frac{1}{2}\right)^2 - \frac{1}{4}} dx$$

let $x + \frac{1}{2} = \frac{1}{2} \sec \theta \rightarrow x = \frac{1}{2} \sec \theta - \frac{1}{2}$

$dx = \frac{1}{2} \sec \theta \tan \theta d\theta$ $x = \frac{1}{2}(\sec \theta - 1)$

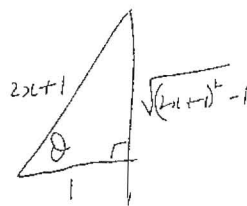
$$= x \ln(\sqrt{x} + \sqrt{1+x}) - \frac{1}{4} \int \frac{(\sec \theta - 1) \sec \theta \tan \theta d\theta}{\sqrt{\frac{1}{4} \sec^2 \theta - \frac{1}{4}}} = \frac{1}{2} \tan \theta$$

$$= x \ln(\sqrt{x} + \sqrt{1+x}) - \frac{1}{4} \int (\sec \theta - 1) \sec \theta d\theta$$

$$= x \ln(\sqrt{x} + \sqrt{1+x}) - \frac{1}{4} \int \sec^2 \theta - \sec \theta d\theta$$

$$= \frac{1}{4} [\tan \theta - \ln(\sec \theta + \tan \theta)] + C$$

Now $\sec \theta = \frac{x + \frac{1}{2}}{\frac{1}{2}} = 2x + 1$ $-\frac{1}{4} \left[\sqrt{(2x+1)^2 - 1} - \ln(2x+1 + \sqrt{(2x+1)^2 - 1}) \right]$



many other forms



$$y = 2(e^{ax} - e^{-ax})$$

$$y^2 = \frac{1}{4}(e^{2ax} - 2 + e^{-2ax})$$

$$y^2 + 1 = \frac{1}{4}(e^{2ax} - 2 + 4 + e^{-2ax})$$

$$y^2 + 1 = \frac{1}{4}(e^{ax} + e^{-ax})^2$$

$$\sqrt{y^2 + 1} = \frac{1}{2}(e^{ax} + e^{-ax})$$

$$\frac{dy}{dx} = \frac{1}{2}(ae^{ax} + ae^{-ax}) = \frac{a}{2}(e^{ax} + e^{-ax})$$

$$y + \sqrt{y^2 + 1} = e^{ax}$$

$$\ln(y + \sqrt{y^2 + 1}) = ax$$

$$x = \frac{1}{a} \ln(y + \sqrt{y^2 + 1})$$

$$(ii) \left(\frac{dy}{dx}\right)^2 = \frac{a^2}{4}(e^{ax} + e^{-ax})^2$$

$$+ a^2 y^2 = -\frac{a^2}{4}(e^{ax} - e^{-ax})^2$$

$$\therefore \left(\frac{dy}{dx}\right)^2 - a^2 y^2 = \frac{a^2}{4}(e^{2ax} + 2 + e^{-2ax})$$

$$= \frac{a^2}{4}(e^{2ax} - 2 + e^{-2ax})$$

$$= \frac{a^2}{4} \cdot 2 + \frac{2a^2}{4} = a^2$$

$$(iii) \left(\frac{dy}{dx}\right)^2 = a^2 y^2 + a^2$$

$$= a^2(y^2 + 1)$$

Note Hence (use (i))

$$\therefore \frac{dy}{dx} = a\sqrt{y^2 + 1}$$

$$\therefore \frac{dx}{dy} = \frac{1}{a\sqrt{y^2 + 1}}$$

$$\therefore x = \int \frac{1}{a\sqrt{y^2 + 1}} dy \quad \text{but } x = \frac{1}{a} \ln(y + \sqrt{y^2 + 1})$$

$$\therefore \int \frac{dy}{\sqrt{y^2 + 1}} = \ln(y + \sqrt{y^2 + 1}) + C$$

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE : $\ln x = \log_e x, \quad x > 0$